

Pawan Mahawar

Analytics & AI Leader | Driving Enterprise ROI through Scalable ML and Generative AI

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CAREER SUMMARY

- Enterprise Analytics and Generative AI leader with 17+ years of experience building and scaling AI ecosystems that convert advanced analytics into measurable business outcomes. Proven track record of delivering revenue growth, marketing ROI uplift, cost optimization, and enterprise transformation across Pharmaceuticals, BFSI, Telecom, Manufacturing, and Technology sectors.
- Experienced in defining multi-year AI strategy, building AI Centers of Excellence, architecting LLM and Agentic AI platforms, and institutionalizing MLOps and Responsible AI governance. Known for combining deep technical architecture expertise with CXO-level influence and value realization ownership.

WORK EXPERIENCE

Dr Reddy's Laboratories

Principal Head, Commercial & Marketing Analytics | Dec 2022 to Present | Hyderabad

Scope: Define and execute enterprise Commercial Analytics, AI, and Generative AI strategy across 20+ brands | Lead centralized AI & GenAI Center of Excellence spanning data science, GenAI engineering, and analytics | Manage ₹8–10 Cr annual AI portfolio with full accountability for ROI, governance, and scale

- **Led, Designed and productionized an Agentic AI framework** using **Lang Graph** for Brand de-growth diagnostics. Built scalable data pipelines ingesting sales, competitor pricing, and internal process metrics (doctor coverage, joint calls, e-detailing) from Google Big Query.
 - Architected and productionized an enterprise Agentic AI platform using Lang Graph, enabling automated diagnostics of brand de-growth across therapeutic portfolios. Built scalable data pipelines ingesting sales, competitor pricing, doctor engagement metrics, and field activity data from Google Big Query, enabling near-real-time decision intelligence for commercial leadership.
 - Designed a cloud-native microservices architecture deployed on Google Cloud Run, containerizing each AI agent using Docker to ensure isolation, scalability, and fault tolerance across distributed AI services.
 - Established enterprise ML Ops pipelines integrating GitHub Actions and Jenkins, enabling automated model training, testing, container builds, and CI/CD-driven deployment of AI services to Cloud Run, significantly reducing model deployment cycles and ensuring production reliability.
 - Engineered a distributed data and inference layer using PostgreSQL and Google Cloud Fire store, enabling horizontal scalability for large-scale brand analytics workloads across multiple geographies.
 - Implemented LLM optimization techniques including prompt caching and token-level cost monitoring using **Lang fuse**, improving system observability, and reducing LLM inference costs across enterprise GenAI workloads.
 - Built an LLM evaluation framework using **RAGAS and LLM-as-a-Judge**, enabling automated evaluation of multi-agent outputs and improving reliability of AI-generated strategic recommendations for business stakeholders.

The platform enabled leadership teams to identify revenue leakages exceeding ₹500 crore and prioritize turnaround strategies, directly contributing to ₹100+ crore revenue recovery across multiple brands.

- **Portfolio Management Agentic AI Solution for better decisioning around the Clinical and Commercial Molecules:** Built multi agent architecture that fetches data from various clinic trial websites, innovator websites, different medical journals and then score the molecules across 32 dimensions using scoring rubric to understand which molecule the organization should invest in before

the innovator patent expires. Created Enterprise grade solution that updates the score every week based on the new information present in the various data bases. \$15 billion worth of molecule investing opportunity was identified using the solution

- **Designed and deployed a RAG + multi-Agent anomaly detection framework** for stockiest billing fraud and outlier behavior across SKU-level data. Engineered feature stores and time-series aggregates stored in Parquet, improving inventory optimization and reducing CFA-level inefficiencies.
- **Deployed an LLM-based Subject Line Open Rate Predictor for HCP engagement.** Optimized inference by exporting trained models to ONNX, enabling faster, cost-efficient deployment within enterprise environments. The solution helped the Open Rate to improve from 12% to 25 % for the email campaigns
- **Designed and deployed a Brand Propensity Recommendation Model** that predicts the most relevant brand to be recommended to each doctor based on prescribing behavior, specialty, historical engagement, and regional trends. Operationalized real-time inference pipelines and automated personalization of e-detailing slide content, dynamically updating promotional material based on model-driven brand recommendations
- **Developed an end-to-end Diet Recommendation Engine for IBS patients**, leveraging predictive modeling and rules-based optimization. Engineered asynchronous APIs and background processing pipelines for inference and personalization, enabling scalable deployment while maintaining clinical responsiveness and system reliability
- **Built and led the Omnichannel Analytics Center of Excellence (COE)** for India business, establishing robust ETL/ELT pipelines, unified customer and campaign data models, and standardized KPI layers across digital channels.
- **Implemented an enterprise Omnichannel Measurement Framework using Synthetic Control** methods with a Generative AI explainability layer, operationalized through reusable pipelines. Delivered ₹20+ crore uplift in marketing ROI by directly influencing senior leadership decision-making.
- **Designed a scalable, production-grade Marketing Mix Modeling (MMM) framework supporting 20+ brands.** Optimized data ingestion, model retraining workflows, and reporting layers—reducing planning cycles by 60% and enabling evidence-based media allocation.
- Designed and Implemented **Multi Touch Attribution Model framework** to measure the incremental fractional credit for each of the digital touchpoints that lead to increase in Prescriptions per Doctor

Neustar (A TransUnion Company)

Manager, Analytics | Mar 2020 to Dec 2022 | Hyderabad

Scope: *Managed \$10M analytics services portfolio | Led 10–12 member delivery team | Owned client P&L, revenue growth, and solution commercialization*

- Owned delivery and financial performance across multiple enterprise accounts
- Expanded portfolio revenue by \$1M annually through value-based pricing and packaging analytics offerings
- Led Marketing Mix Modeling and media optimization programs influencing multimillion-dollar media investments
- Built repeatable solution frameworks transitioning from project-based delivery to scalable products
- Served as primary CXO partner aligning analytics investments to business outcomes

Cognizant

Deputy General Manager | Feb 2018 to Mar 2020 | Hyderabad

Scope: *Directed analytics strategy across large insurance and healthcare enterprises | Led 20+ consultants across multi-program delivery | Managed multimillion-dollar transformation budgets*

- Led enterprise analytics transformation programs embedding predictive intelligence across customer lifecycle and claims operations

- Delivered \$300M+ claim cost reduction and \$95M+ revenue/premium impact across portfolio
- Standardized delivery operating model and reusable IP to improve scalability across clients
- Acted as senior client advisor influencing roadmap and strategic priorities

Nuevora Analytics

Senior Manager, Banking Vertical | Jun 2016 to Jan 2018 | Hyderabad

Scope: Established onsite analytics practice | Built and led 6–8 member team | Responsible for delivery, stakeholder management, and business value realization

- Set up and scaled analytics practice for Union Bank of Philippines
- Delivered \$750K+ measurable value via behavioral scorecards, CLV, and recommendation systems
- Led real-time optimization programs doubling CTR and improving marketing ROI

EARLIER ROLES

- **American Express | Assistant Manager** | Jun 2013 to Oct 2015 | Gurgaon
- **HSBC – Business Reengineering Consultant** | Apr 2011 to May 2013 | Kolkata
- **Ashok Leyland Ltd – Deputy Manager** | Jul 2006 to May 2009 | Hosur

SKILLS

- **AI & Generative Systems:** Agentic AI | RAG Architectures | LLM Copilots | LangChain | LLM Evaluation Frameworks | Responsible AI
- **Machine Learning:** Predictive Modeling | Marketing Mix Modeling | Causal ML | Forecasting | XGBoost | TensorFlow | PyTorch
- **Cloud & Engineering:** GCP Vertex AI | BigQuery | Python | SQL | FastAPI | Production Pipelines
- **Commercial Analytics:** ROI Modeling | Omnichannel Analytics | Customer Analytics | Incrementality & Lift Measurement
- **Leadership:** Analytics COE Setup | Team Building | Budget Ownership | Stakeholder Management | CXO Partnership | AI Strategy & Roadmaps

CERTIFICATIONS

- AI & ML Certified – IIIT Hyderabad
- Master Black Belt – ISI Mumbai
- US CFA Level 2 Candidate

EDUCATION

- MBA – Narsee Monjee Institute of Management Studies, Mumbai – Jun 2009 to Apr 2011
- Bachelor of Mechanical Engineering – NIT Trichy – Aug 2002 to May 2006